Rolling Class Software Setup: Common Steps for Ubuntu and MAC OS

Big Data Algorithms — setup guide

Before you start this guide, you should have completed either "02_software_setup_ubuntu.qmd" (for Ubuntu, including Ubuntu inside WSL on Windows) or "02_software_setup_mac.qmd" (for a Mac). Those guides install R, Python, PostgreSQL, Quarto and Positron. This guide installs the R and Python packages we use in the course and sets up Quarto to run R and Python in the same document. From here on, the steps are the same on Ubuntu and on a Mac.

Convenience Setup

Before we continue, let's set up a convenience: we would like to be able to run lines of bash (linux) in the linux terminal directly from the `qmd` file currently open.

Assuming you're using Positron, we need the ability to send commands from this `qmd` document directly to terminal. (By default this is only enabled for R or Python). So follow these instructions:

Open File → Preferences → Keyboard Shortcuts, or press Ctrl+K and then Ctrl+S. In the search box, type "Run Selected Text In Active Terminal". Click the pencil icon next to that command, press the keys you want (for example Ctrl+Alt+Enter), then press Enter. You don't want to use Ctrl+Enter as this will conflict with R/Python later.

Run every command in the terminal. A few commands are too long to fit on one line, so they continue on the following lines, either after a \ or inside quotes. Copy all the lines together and the terminal runs them as one command.

Python environment and packages

We will add packages to this guide during the semester. The last section explains what to do when that happens.

Neither Ubuntu nor Homebrew allows `pip` to install packages into the Python they manage (you would get an `externally-managed-environment` error). Instead, create a virtual environment for the course and install packages there:

```
python3 -m venv ~/.venvs/bigdata
source ~/.venvs/bigdata/bin/activate
```

Your prompt now begins with `(bigdata)`, and inside the environment the commands `python` and `pip` refer to the environment's own copies. Type `deactivate` to leave it. You need to run the `source` line again in every new terminal where you want to use the environment. Positron looks for environments in `~/.venvs`, so it will find this one on its own.

With the environment active, install the Python packages we use in the course, so that everyone works with the same set:

```
pip install numpy pandas polars duckdb pyarrow scikit-learn lightgbm \
  statsmodels requests ipykernel "psycopg[binary]" "ibis-framework[postgres]" \
  "xgboost-cpu; sys_platform == 'linux'" \
  "xgboost; sys_platform == 'darwin'"
```

`psycopg` is the PostgreSQL driver for Python (the `[binary]` part, quotes included, selects a ready-built version so nothing needs compiling). `ibis` writes the SQL for you from Python code, and its `[postgres]` part adds what it needs to talk to PostgreSQL. `statsmodels` fits classical statistical models (regressions with standard errors and p-values), and `requests` downloads data from web pages and APIs. The last two entries, quotes included, are two versions of the same package, and `pip` picks the one for your system. On Ubuntu it installs `xgboost-cpu`, a build without GPU support; the regular build pulls in several hundred megabytes of NVIDIA libraries that only matter when training on multiple GPUs. On a Mac, where no such build exists or is needed, it installs the regular `xgboost`. Either way the package is used as `xgboost` in your code. Near the start of its output, `pip` prints a line beginning with `Ignoring` about the version it did not install, which is expected.

This downloads a couple of hundred megabytes, so it can take several minutes. Check that the packages load:

```
python -c "import numpy, pandas, polars, duckdb, pyarrow, psycopg, ibis, statsmodels, request
python -c "import sklearn, xgboost, lightgbm; print('ok')"
```

Both commands should print `ok`.

R packages

Install the R packages we use in the course:

```
Rscript -e 'install.packages(c("tidyverse", "data.table", "R.utils",
  "arrow", "duckdb", "RPostgres", "bench", "profvis", "httr2", "digest",
  "future", "furrr", "xgboost", "lightgbm",
  "knitr", "rmarkdown", "reticulate"))'
```

The packages listed need many others, so well over a hundred packages are installed and it takes several minutes. When one package fails, R moves on to the next and only prints a warning at the end, so check that the main ones load:

```
Rscript -e 'suppressMessages({library(tidyverse); library(data.table)
  library(arrow); library(duckdb); library(RPostgres); library(xgboost)
  library(lightgbm); library(reticulate)})
  cat("ok\n")'
```

This should print `ok`. If it says `there is no package called` something, look through the output of the install command for a line saying `installation of package ... had non-zero exit status`. That package had to be compiled on your computer and something it needs was missing, almost always a system library. On Ubuntu, the usual cause is having skipped the `sudo apt install` line in the "Prepare R for installing packages" section of the Ubuntu guide; `RPostgres`, for example, needs `libpq-dev` from that line. Run that line, then run the `install.packages` command above again.

Connecting to PostgreSQL from R and Python

In the earlier guide you saw that `psql` connects without a password. The same is true from R and Python, with one rule: **do not pass a `host`**. Check both:

```
Rscript -e 'con <- DBI::dbConnect(RPostgres::Postgres()); print(DBI::dbGetQuery(con, "SELECT
python -c "import psycopg; print(psycopg.connect().execute('SELECT version()').fetchone()[0]
```

Each should print a line beginning with `PostgreSQL 18`.

Why the rule: with no `host`, the connection goes through a local socket file and PostgreSQL identifies you by your operating-system username, which is why no password is needed. With `host = "localhost"`, which many tutorials show, the connection goes over the network instead, and PostgreSQL then insists on a password that your setup never created. The

error is `fe_sendauth: no password supplied`. If you ever do need a network connection (for example from a program that cannot use the socket), set a password once with `\password` inside `psql` and then pass `host = "localhost"` together with `password`.

R and Python together in Quarto

A single Quarto document can contain both R and Python code. Quarto runs the code in such a document with R's `knitr` and `rmarkdown` packages, and the `reticulate` package lets R start Python for the Python parts. All three were installed in the previous section.

Left alone, `reticulate` chooses a Python by itself, and it may not be your `bigdata` environment. Packages you installed there would then appear to be missing when you render a document. Tell `reticulate` which Python to use by setting the variable `RETICULATE_PYTHON` in `~/.Renviron`, a file R reads every time it starts:

```
echo "RETICULATE_PYTHON=$HOME/.venvs/bigdata/bin/python" >> ~/.Renviron
```

Check the result:

```
Rscript -e 'reticulate::py_config()'
```

The first line should read `python:` followed by the path of `.venvs/bigdata/bin/python` inside your home directory. Further down, the `numpy:` line should show a folder inside `.venvs/bigdata` as well, which confirms that R sees the Python packages you installed above.

Ensure the setup is correct

Let's make sure you have R and Python setup correctly. Here are two chunks that should test it. Go to console on the bottom pane and click the "+" sign to start both an R session. Then click the down icon next to plus and click "start another" and choose the python in

```
7 + 3 #CTRL+ENTER or click "run cell"
```

```
[1] 10
```

```
7 + 3 #CTRL+ENTER or click "run cell"
```

```
10
```

When the package lists change

When we add a package to one of the lists above, repeat that step:

- For Python, activate the environment with `source ~/.venvs/bigdata/bin/activate` and run the `pip install` command again. `pip` skips the packages you already have.
- For R, run the `install.packages` command again. R installs all the listed packages again, which also brings them up to date.